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Python Programming Foundations

Week 1 - First Programs and Basic Values
Day 3: "Basic Values Together"

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Put The Pieces Together

This week, you learned the smallest useful Python pieces.

Today, you combine them into tiny programs that run and make sense.

Week 1 Python Concepts Working Together
These ideas combine to create a small program.

Small pieces can make a useful script.

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Quick Review

Monday made output visible.

Tuesday made numeric value flow visible.

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What Counts as Success Today

NOT a large program

Small Program that:

- Runs
- Produces output
- Can be Explained

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What We Will Use Today

- ▶ strings
- ▶ numbers
- ▶ variables
- ▶ simple expressions
- ▶ `print()`
- ▶ optional `input()` if stable
- ▶ basic type awareness

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Day 3 Working Set

The simple tools you will use today.

Use the pieces you can explain.

Working Set

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Topics Parked for Later

We are focusing on the basics right now.

Later

- lists**
Store ordered collections of items.
- dictionaries**
Store key-value pairs for fast lookups.
- loops**
Repeat steps many times.
- functions**
Group steps and reuse them.
- files**
Read from and write to the computer.
- AI-generated code**
Powerful help in the future.

These are future tools, not today's requirements.

What to "Skip" for Now

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A Small Program Can Still Be Useful

Tiny scripts solve real, everyday problems.

Input / Start Values

```
miles = 3
rate = 0.65
```

Program Calculates

Output

```
Reimbursement: 1.95
```

Calculation: `miles * rate = reimbursement`

Useful does not have to mean large.

A useful first program does not have to be large.
It only needs a clear purpose, visible output, and values you can explain.

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Combine Values Carefully

Your program may use text and numbers together.

The important question is whether you can explain each value's role.

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Type Awareness Through Use

For Week 1, recognize:

- ▶ Text
- ▶ Whole numbers
- ▶ Decimal numbers
- ▶ True/false values (boolean)

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Basic Type Awareness Through Use

Python works with different kinds of values.

Text	Whole number	Decimal number	True/False
"Riley"	2	87.5	True
Words and text in quotes.	Counted items with no decimal.	Numbers that include a decimal part.	Values that are either True or False.

Different values can behave differently in code.

Type Awareness

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Explain Every Main Line

Before submitting Assignment 1, ask: can I explain every main line?

If a line works but you cannot explain it, slow down and inspect it.

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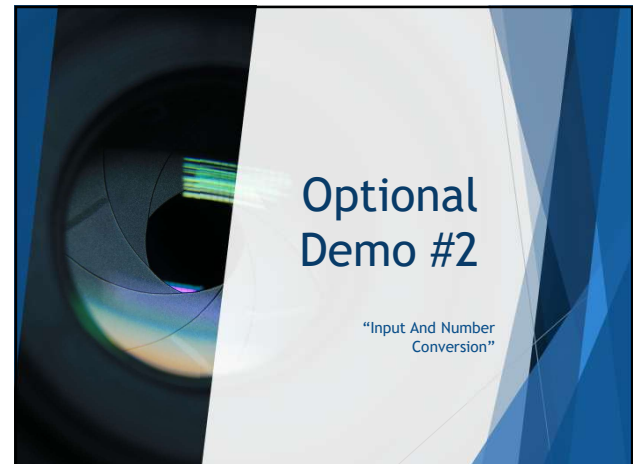
 A slide titled "Basic Type Awareness Demo" with the Southwest Tech logo. It includes a table showing variable names, values, and kinds of values, and a note about recognizing values before explaining the program.

Variable Name	Value	Kind of Value
student_name	"Riley"	text
assignments_completed	2	whole number
average_score	87.5	decimal number
is_passing	True	true/false

Recognize the value before you explain the program.

Type Demo

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 A slide titled "Optional Demo: Input And Number Conversion" with the Southwest Tech logo. The background shows a close-up of a computer keyboard. The text says:

- ▶ When users type into `input()`, Python receives text first.
- ▶ For math, the text must be converted into a number.

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Learning Move: *Type It When It Is New*

Typing code can be part of learning.

Copying is not automatically wrong, but it should not replace understanding.

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Typing it Out

Typing code manually slows the process down in a useful way.

When you type an example, you are not just moving characters onto the screen. You are giving your mind time to process:

- ▶ the order of the instructions
- ▶ the names of variables
- ▶ the structure of the logic
- ▶ the punctuation and syntax
- ▶ the relationship between one line and the next
- ▶ what the program is actually doing

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Copying and Pasting

Copying and pasting can also be appropriate. It may make sense when:

- ▶ the code is long
- ▶ the concept has already been learned
- ▶ the focus is on a different part of the task
- ▶ you understand the logic already
- ▶ the code is being used as a scaffold for something else

Copy/paste becomes a problem when it replaces understanding. - Before copying, You should ask:

- ▶ Do I understand what this code does?
- ▶ Do I need to understand this more deeply right now?
- ▶ Am I copying because it saves time, or because I am avoiding thinking?

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Connection to AI Prompting

Typing a prompt is not just the act of entering text into a tool. It is often part of the thinking process.

As a person writes a prompt, they may:

- ▶ clarify the goal
- ▶ revise wording
- ▶ discover missing constraints
- ▶ notice ambiguity
- ▶ reshape the request
- ▶ improve the structure of their own thinking

This is closely connected to AI-assisted development and Refraction-Based Architecture.

- ▶ The human is not merely asking AI for output - *The human is shaping intent!*

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Typing vs. Copying: Use Both Wisely

Two useful paths for building your skills.

Type When Learning



slows you down enough to notice

vs.

Copy When Understood



useful when the logic is already clear

 The goal is understanding, not typing forever.

Typing vs Copy-Pasting

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Common Failure: Too Much Is Too Much



Trying to impress can make the first assignment harder than it needs to be.



Small, clean, and explainable is better than large and confusing.

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Common Failure: Working But Unexplained

- ▶ Code that runs is not the whole goal.
- ▶ You also need to explain what values the program uses and what output it creates.

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From Demo to Lab

For Assignment 1:

- ▶ Finish two or three tiny programs.
- ▶ Each program should run, show output, and make sense when you explain it.

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Assignment 1: Completed

Three small Python programs. Run them. Understand them.

 greeting.py	✓	Output: Hello, Riley!
 calculator.py	✓	Output: 7 + 5 = 12
 converter.py	✓	Output: 3 miles = 1.95

★ Two or three tiny programs that run and make sense.

Assignment Completion

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Assignment 1: Beginner Evidence

Show that your programs run and make sense.

.py files	readable output	short explanation note
- greeting.py - calculator.py - converter.py	<pre>Hello, Riley! 7 + 5 = 12 3 miles = 1.95</pre>	This program uses starting values, calculates a result, and prints the output.

 +  +  = Code + output + explanation

Assignment Evidence

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Lab On Success Check

- ▶ Week 1 worked if your small programs run and you can explain them.
- ▶ Understanding what the code is doing:
 - What values does (each) program use?
 - What output does it produce?
 - Where does a value change?
 - Could you explain this script line by line to someone else?

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Thank you!
Have Fun!

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